

The Constants of Nature

$$\begin{aligned}
d_{cs} &= \frac{E_1}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_n}{\mu_e} &= -\frac{S}{C_{\text{CFP}}} \left(\frac{m_n}{m_e} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_e^2 \right) \boxtimes \\
\text{eV} : \text{Hz} &= \frac{eV}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_n}{\mu_e} &= -\frac{C_{\text{CFP}}}{5} \left(\frac{m_n}{m_e} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_e^2 \right)^{-1} \boxtimes \\
\text{J} : \text{Hz} &= \frac{\text{joule}}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_n}{\mu_e} &= \frac{C_{\text{CFP}}}{P_{\text{up}}} \left(\frac{m_n}{m_e} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right) \boxtimes \\
\text{Hz} : \text{J} &= 2\pi \left(\frac{t_p m_p}{s l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_n}{\mu_e} &= \frac{C_{\text{CFP}}}{m_e} \left(\frac{m_n}{m_e} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_e^2 \right)^{-1} \boxtimes \\
\text{Hz} : \text{eV} &= \frac{2\pi}{eV} \left(\frac{t_p m_p}{s l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & A_{\text{mdm}} &= \mathcal{K}_5 \left(\frac{t_p^2 q_p m_p}{l_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
h &= \frac{1}{eV} \left(\frac{t_p^2 m_p}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \mu_\theta &= \frac{\mathcal{K}_5}{2} \left(\frac{t_p^2 q_p m_p}{l_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
h &= \left(\frac{t_p m_p}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \mu_N &= \frac{\mathcal{K}_5}{2} \left(\frac{t_p^2 q_p m_p}{l_p m_n} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
h &= 2\pi \left(\frac{t_p^2 m_p}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \mu_{\text{tri}} &= \mathcal{K}_5^2 \mathcal{K}_2^2 \left(D_{\text{D0}} + \pi \right) \left(\frac{t_p^2 q_p m_p}{l_p} \right) \left(1 - \omega_2 \left(\frac{2\pi}{18} \right) \mathcal{K}_e^2 \right) \boxtimes \\
q_e &= \pi \left(\frac{t_p^2 m_p}{l_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_{\text{tri}}}{\mu_e} &= S \left(D_{\text{D0}} + \pi \right) \left(\frac{m_n}{m_e} \right) \left(1 + \frac{4}{L_{\text{Li}}} \left(\frac{2\pi}{4\pi} \right) \mathcal{K}_e^2 \right) \boxtimes \\
2q_c &= 2\pi \left(\frac{t_p^2 m_p}{l_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_{\text{tri}}}{\mu_n} &= \mathcal{K}_5 \mathcal{K}_2^2 \left(2D_{\text{D0}} + 2\pi \right) \left(\frac{m_n}{m_e} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
\mu_\theta &= \frac{1}{2} \left(\frac{t_p^2 C m_p}{l_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_{\text{tri}}}{\mu_\theta} &= \mathcal{K}_5 \mathcal{K}_2^2 \left(2D_{\text{D0}} + 2\pi \right) \left(\frac{m_n}{m_e} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
\mu_N &= \frac{1}{2} \left(\frac{t_p^2 C m_p}{l_p m_n} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_0^2 \right) \boxtimes & g_{\text{tri}} &= \mathcal{K}_5 \mathcal{K}_2^2 \left(4D_{\text{D0}} + 4\pi \right) \left(\frac{m_n}{m_e} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^2 \right) \boxtimes \\
G_0^{-1} &= \frac{\pi}{\mathcal{K}_5^2} \left(\frac{t_p^2 m_p}{l_p q_p} \right) \left(1 + \sqrt{\zeta(3)} \, \mathcal{K}_0^2 \right) \boxtimes & \mu_e &= \left(\frac{m_e}{m_e} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_e^2 \right) \boxtimes \\
G_0 &= \frac{\mathcal{K}_5^2}{\pi} \left(\frac{t_p q_p^2}{l_p m_p} \right) \left(1 - \sqrt{\zeta(3)} \, \mathcal{K}_0^2 \right) \boxtimes & G &= \left(\frac{t_p^3}{l_p^3 m_p} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_e^3 \right) \boxtimes \\
\text{eV} &= \mathcal{K}_5 \text{joule} \left(\frac{q_p}{C} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & G &= \text{GeV}^2 \left(\frac{t_p^4}{l_p^4 m_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_e^2 \right) \boxtimes \\
\text{J} : \text{eV} &= \frac{1}{\mathcal{K}_5} \left(\frac{C}{q_p} \right) \left(1 - \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & m_n &= \left(\frac{2\pi}{25} \right)^2 \frac{\text{GeV}}{\varrho^{-2}} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \left(\frac{2s}{4\pi} \right) \mathcal{K}_e^3 \right) \boxtimes \\
-\frac{e}{m_e} &= -\mathcal{K}_5 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & m_+ &= \left(\frac{2\pi}{s} \right)^2 \frac{\text{GeV}}{\omega_2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \frac{\sqrt{2}}{e^{\gamma/3}} \left(\frac{18}{25} \right) \mathcal{K}_e^3 \right) \boxtimes \\
\frac{e}{m_+} &= \mathcal{K}_5 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_{de}}{\mu_e} &= -\rho_{\text{up}} \sqrt{\frac{2s}{6\pi}} \left(\frac{m_e}{m_{de}} \right) \left(1 - 18 \sqrt{L_1} \mathcal{K}_e^3 \right) \boxtimes \\
\frac{h}{h} &= \frac{\mathcal{K}_5}{4\pi} \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_e}{\mu_{de}} &= -\frac{1}{2\pi} \sqrt{\frac{6\pi}{2s}} \left(\frac{m_{de}}{m_e} \right) \left(1 - 18 \sqrt{L_1} \mathcal{K}_e^3 \right)^{-1} \boxtimes \\
\frac{H_N}{h} &= \frac{\mathcal{K}_5}{4\pi} \text{MHz} \left(\frac{q_p}{s m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{K}_0^2 \right) \boxtimes & m_p &= \frac{2\pi}{35} \sqrt{\frac{4\pi}{6s}} \left(A_{\text{mass}} \right) \left(1 - 18 K_{-6} \mathcal{K}_e^3 \right) \boxtimes \\
a &= \frac{4\pi}{32 s \mathcal{K}_5^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{K}_0^2 \right) \boxtimes & m_{de} &= \frac{4\pi}{\sqrt{35}} \left(\frac{6\pi}{4s} \right) \left(A_{\text{mass}} \right) \left(1 + K_{-3}^5 \left(\frac{7\pi}{9s} \right) \mathcal{K}_e^3 \right) \boxtimes \\
d_{220} &= \frac{1}{\sqrt{8}} \frac{4\pi}{32 s \mathcal{K}_5^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{K}_0^2 \right) \boxtimes & \frac{\mu_{de}}{\mu_n} &= S \sqrt{\frac{2s}{6\pi}} \left(\frac{m_n}{m_{de}} \right) \left(1 - \frac{15}{\varrho} \left(\frac{4\pi}{35} \right) \mathcal{K}_e^3 \right) \boxtimes \\
A_{90} &= \text{ampere} \left(1 + \frac{4}{s} \left(\frac{4\pi}{35} \right) \mathcal{K}_0^3 \right) \boxtimes & \frac{\mu_{de}}{\mu_n} &= -C_{\text{CFP}} \sqrt{\frac{2s}{6\pi}} \left(\frac{m_n}{m_{de}} \right) \left(1 - 35 \text{Im}(\omega_1) \left(\frac{4\pi}{32} \right) \mathcal{K}_e^3 \right) \boxtimes \\
C_{90} &= \text{coulomb} \left(1 + \frac{4}{s} \left(\frac{4\pi}{35} \right) \mathcal{K}_0^3 \right) \boxtimes & \mu_{he} &= \frac{\mathcal{K}_5}{4\pi} \left(\frac{t_p q_p}{l_p m_e} \right) \left(1 + s \left(\frac{K}{6} \right) \mathcal{K}_0^3 \right) \boxtimes \\
\frac{\mu_\theta}{hc} &= \frac{\mathcal{K}_5}{4\pi} \left(\frac{t_p q_p}{l_p m_e} \right) \left(1 + s \left(\frac{K}{6} \right) \mathcal{K}_0^3 \right) \boxtimes & \mu_{he}' &= -\frac{1}{2} \sqrt{\frac{1}{4\pi}} \left(\frac{1}{4s$$

$t_p = 5.39125836832313 \times 10^{-38}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-35}$ m	Planck length
$t_p = 1.87554596713962 \times 10^{-38}$ K	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.1766268317579 \times 10^{-27}$ kg	Planck mass
$\boxtimes = 0.199499199973626 \times 10^{-10}$ hyperbolic inversion boundary	
$\boxtimes = 0.074997112207 \times 10^{-10}$ number of dimensions	
$\boxtimes = 4.48726242916751$ break in scale symmetry	
$\eta_{k1} = 0.08542454315533004 \dots$	1 st hyperbolic partition constant
$\eta_{k2} = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\eta_{k3} = 1.87646903900417 \times 10^{-4} + 4.06615262615972 \times 10^{-4}$	3 rd hpc
$\eta_{k4} = 1.87646903900417 \times 10^{-4} - 4.06615262615972 \times 10^{-4}$	4 th hpc
$\eta_{k5} = 4.48726242916751$	hyperbolic radius constant
$\eta_{k6} = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{60} = 1.01494160640965 \dots$	Gieseking's constant
$V_{60} = 0.20988321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.9159655941727190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_d = 1.72616281206959 \dots$	dimer constant
$\omega_1 = 0.7649791818528856 + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.529954037057192 \dots$	omega_2 constant
$t^{(1)} = 0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$\varphi_i = (-1)^{1/2}$	imaginary unit
$\varphi = 1.61803398874989 \dots$	imaginary golden ratio
$\pi = 3.14159265358979 \dots$	golden ratio
$P_{60} = 2.295587514939263 \dots$	Archimedes' constant
$S = 5.244115110858423 \dots$	universal parabolic constant
$S = 2.622057554292119 \dots$	arc length of the unit lemniscate
$L_1 = 1.31102877174605 \dots$	lemniscate constant
$L_2 = 0.599070117367796 \dots$	1 st lemniscate constant
$G_0 = 0.847213084793979 \dots$	2 nd lemniscate constant
$G_{60} = 0.834626841674073 \dots$	ubiquitous constant
$W_{60} = 0.4749493799879109 \dots$	Gauss's constant
$C_{60} = 2.2028255576595772 \dots$	Weierstrass constant
$\eta_{k1} = 0.9981360445985020 \dots$	Ramanujan's 1 st continued fraction constant
$M = 0.1100010000000000 \dots$	Liouville's constant
$M = -1.74756459463318 \dots$	Madelung constant
$P = 1.226742010172035 \dots$	prime constant
$e = 2.71828182845904 \dots$	Euler's number
$x_{60} = 2.29316628741186 \dots$	1 st Foias constant
$\gamma = 0.57725664901532 \dots$	Euler-Mascheroni constant
$S = 0.828225429678840 \dots$	Sierpiński's constant
$D_{60} = 0.739085133215160 \dots$	Dottie number
$L_{60} = 0.662743419349181 \dots$	Laplace limit
$C_{60} = 1.19967864025773 \dots$	real fixed point of the hyperbolic cotangent
$J_{01} = 2.40284255576595772 \dots$	1 st root of the Bessel function
$\alpha_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\alpha_F = 0.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.246979603717114 \dots$	silver constant
$P = 0.6247135724474 \dots$	plastic constant
$C = 1.3054439657196732 \dots$	QRS constant
$K_{60} = 1.15655237442151 \dots$	Khinchin mean of order -6
$T_0 = 2.731500000000000 \dots \times 10^5$ pascal	freezing point of water
$P_0 = 1.000000000000000 \dots \times 10^2$ kelvin	1 bar
$P_1 = 1.013250000000000 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules	energy associated with ΔE_{cs}
$\nu_1 = 5.400000000000000 \dots \times 10^{14}$ Hz	frequency of max/light conversion
$\nu_{peak} = 4.965114237174427 \dots$	peak 2 of spectral radiant unit wavelength
$\nu_{peak} = 2.82143939712107 \dots$	peak emission of spectral flux/unit frequency
$n!$ = factorial function	
$\ln$ = derangement function	
$\log(x)$ = natural logarithm	$lm = \text{lumen} = \text{cd} \cdot \text{sr}$
$\Gamma(x)$ = gamma function	$N = \text{newton} = \text{m kg/s}^2$
$\zeta(x)$ = Riemann zeta function	$\mathcal{N} = \text{noether} = \text{m kg/s}^2$
	$\text{ol} = \text{ohm} = \text{m}^2 \text{ kg/s}^2 \text{ C}^2$
units	$\text{Pa} = \text{pascal} = \text{kg/m}^2 \text{ s}^2$
m = meter	$S = \text{siemens} = \text{s}^2 \text{ C}^2/\text{m}^2 \text{ kg}$
s = second	$T = \text{tesla} = \text{kg/m}^2 \text{ s}^2 \text{ C}$
C = coulomb	$V = \text{volt} = \text{m}^2 \text{ kg/s}^2 \text{ C}$
K = kelvin	$W = \text{watt} = \text{m}^2 \text{ kg/s}^2$
$\text{kg} = \text{kilogram}$	$\text{Wb} = \text{weber} = \text{m}^2 \text{ kg/s}^2 \text{ C}$
$\text{J} = \text{joule} = \text{m}^2 \text{ kg/s}^2$	
derived units	
MHz = megahertz = $1.00000000000000 \dots \times 10^6$ 1/s	
fm = femtometer = $1.00000000000000 \dots \times 10^{-15}$ m	
eV = electron-volt = $1.00000000000000 \dots$ eV	
MeV = megaelectron-volt = $1.00000000000000 \dots \times 10^6$ eV	
GeV = gigaelectron-volt = $1.00000000000000 \dots \times 10^9$ eV	
$\frac{r_n}{r_1} = \left(\frac{4\pi}{5}\right)^2$	$\frac{r_n}{r_1} = K$
$\frac{r_n}{r_1} = \$	